

Software System Design (2024) - Architecture

Assignment 1

(20 Points)

You are responsible to INDEPENDENTLY complete three individual tasks related to software quality attributes and tactics, then compile and submit your report.

DEADLINE: 12:00am on (Monday) May 13, 2024

Task 1: Quality Attribute Scenarios (8 points)

With the rapid development of Large Language Model (LLM), soon it will be an enabling component embedded in many contemporary software systems. Do your own research, and figure out **FOUR quality attributes** with the highest priority that such software systems have to achieve, at least **TWO of which are NOT introduced in lectures**. Apply scenario-based (stimulus-response) analysis method to create their **general scenarios** and typical **concrete scenarios** (using six-element analysis with at least **THREE response measures**) respectively based on your research and understanding. Present the general scenario in the format of six-element table and the concrete scenario in the stimulus-response diagram.

Task 2: Strategies and Tactics (6 points)

Discuss at least TWO possible **strategies** to improve each of the above quality attributes; propose at least TWO different **tactics** for each possible strategy and identify their potential **impacts on other attributes**. Use a table with columns of quality attributes and rows of strategies and tactics to discuss the benefits and the penalties of the potential impacts that may come with each tactic.

Task 3: Alternative Tactic Debate (6 points)

Enumerate the possible benefits and the costs of 'active redundancy', 'passive redundancy' and 'spare' (not covered in Task 2) — the alternative tactics introduced for Availability in lecture. Discuss at least **THREE different scenarios** that any of them may better fit in respectively with your reasons behind. Conclude the **criteria** for how to make the right choice among them, and elaborate the **rationale** supporting your argument.

Assignment Deliverable:

The assignment deliverables should be submitted individually. Each deliverable contains no more than 10 pages (A4 size) plus a cover page. Note your individual submission will be plagiarism-checked. You'll receive no point if plagiarism is detected in any task of your assignment with others.